SaaS Churn & Revenue Analytics

What drives churn, support load and revenue at RavenStack — a synthetic SaaS startup — answered from 5 linked tables: 500 accounts, 4,996 subscriptions, 25,000 feature-usage events, 2,000 support tickets and 600 churn events.

Prepared by: Piyush Saxena | **Programme:** Future Interns — Data Science & Analytics, Task 2 | **Dataset:** RavenStack Synthetic SaaS Dataset by River @ Rivalytics (MIT-like, fully synthetic, no PII) | **Tools:** Python (pandas, matplotlib), Microsoft Excel

500	22.0%	$11.33M	600
ACCOUNTS 2023–2024 signups	ACCOUNT CHURN 110 of 500 accounts	RECORDED MRR (CLEANED) 4,996 subscription records	CHURN EVENTS 352 unique accounts
25,000	**2,000**	**3.98 / 5**	**35.9 hrs**
FEATURE-USAGE EVENTS 40 features	SUPPORT TICKETS 98% of accounts raised one	AVG SATISFACTION 1,175 responses (58.8%)	AVG TICKET RESOLUTION Urgent barely faster than low

EXECUTIVE SUMMARY

RavenStack's churn problem is **not where most teams would look for it**. Plan tier, trial status, billing frequency, auto-renew, seat count, support-ticket volume and even product usage separate churned and retained accounts by almost nothing — in this data they are near-identical. Two things **do** separate them: **industry** (DevTools churns at 31.0%, almost double Cybersecurity's 16.0%) and **how the customer arrived** (event-acquired accounts churn at 30.2%; partner-acquired at just 14.6%). Revenue, meanwhile, is dangerously concentrated: Enterprise is only 34% of subscription records but 75% of recorded MRR. And support has a hidden problem — urgent tickets are resolved barely faster than low-priority ones.

DATA QUALITY & DUPLICATE CHECK — DONE FIRST, BEFORE ANY ANALYSIS

On the intern's instruction, every table was checked for duplicates and integrity **before** analysis. Full-row duplicates: **zero in all 5 tables**. Primary keys are unique in accounts, subscriptions, support_tickets and churn_events. Foreign keys are perfect — **zero orphan rows** in any table. Three issues were found and handled explicitly:

- **feature_usage — usage_id is NOT unique:** 21 usage_id values are each used twice, but for completely different events (different subscription, date and feature) — 42 rows involved. These are ID collisions from data generation, not duplicate rows, so **no rows were deleted**; each row is treated as a separate event and usage_id is not used as a key.
- **subscriptions — 4 true duplicate pairs:** 8 rows form 4 pairs that are identical in every field except subscription_id (same account, start date, plan, seats and MRR — all starting in Dec 2024). Keeping both would double-count $8,544 of MRR, so **one copy of each pair was removed: 5,000 → 4,996 records**. All revenue figures in this report use the cleaned 4,996.
- **Expected nulls (not errors):** subscriptions.end_date is null for 4,510 cleaned records (90.3%) = still-active subscriptions; support satisfaction_score is null for 825 tickets (41.3%) = customer never responded; churn feedback_text is null for 148 events (24.7%) = no comment left. Multiple churn events per account (600 events, 352 accounts) are legitimate — accounts churn, reactivate (10.2% of events) and may churn again.

A key reconciliation warning: accounts.churn_flag marks only 110 accounts (22.0%) as churned, yet churn_events log 352 unique accounts — because most of those accounts reactivated and are active again at the snapshot. The two numbers measure different things and must never be reported as one 'churn rate'.

1 WHO CHURNS — AND WHO DOESN'T

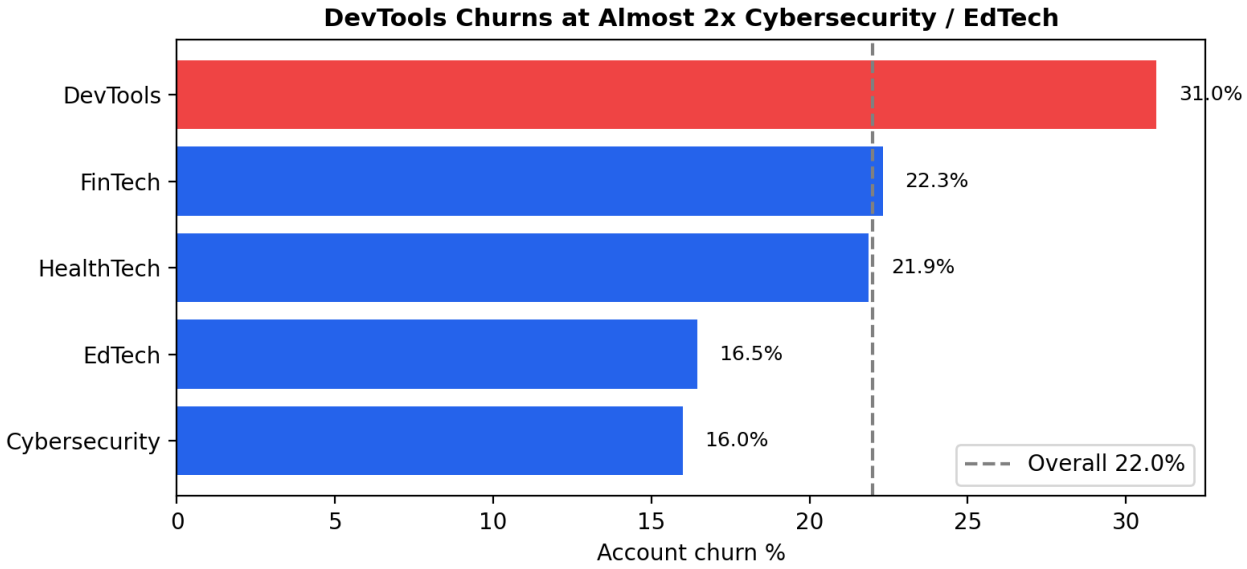


Figure 1 — Account churn by industry. DevTools (31.0%) is the outlier.

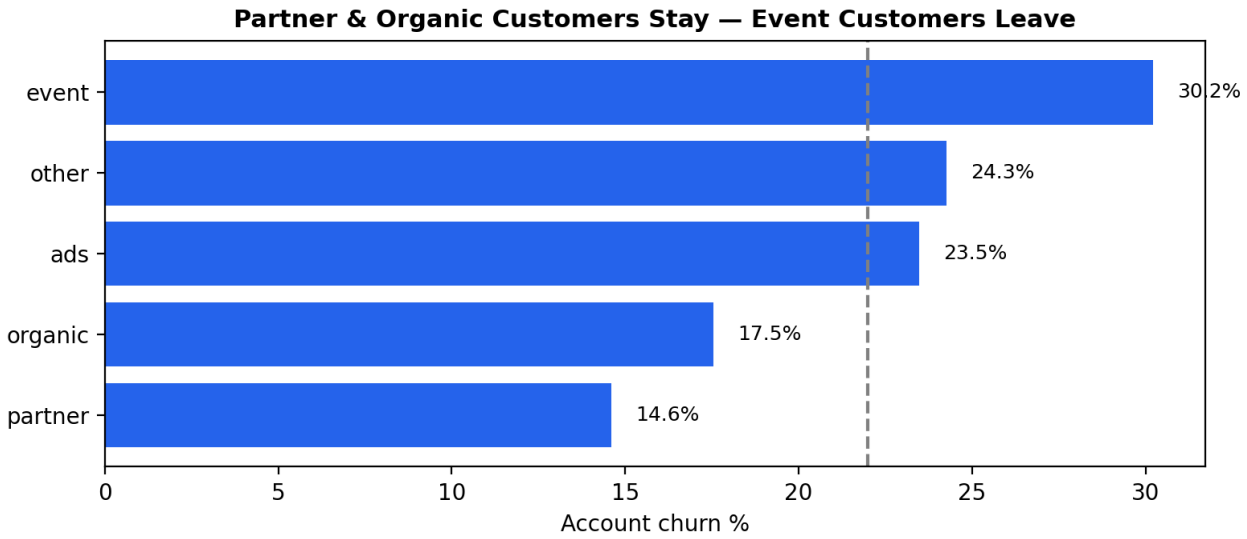


Figure 2 — Account churn by referral source. Channel quality differs by 2x.

What was tested	Churned	Retained	Verdict
Plan tier (Basic / Pro / Enterprise)	21.9–22.1%	—	**No effect** — all tiers churn equally
Trial accounts	25.8% churn	21.1% non-trial	Small effect only
Subscription trial / billing / auto-renew	9.4–10.0%	—	**No effect**
Seats (avg)	19.2 seats	20.9 seats	Almost no difference
Support tickets per account (avg)	3.93	4.02	**No effect**
Total product usage (avg)	521.6	494.7	Churned accounts used *more*
Industry	DevTools 31.0%	Cybersecurity 16.0%	**Strong effect**
Referral source	Event 30.2%	Partner 14.6%	**Strong effect**

This table is the most honest page of the report: for a synthetic dataset many classic churn drivers are deliberately weak, and saying so — instead of inventing a story — is the correct analyst call. The actionable signals are industry and acquisition channel.

2 REVENUE — CONCENTRATED IN ENTERPRISE

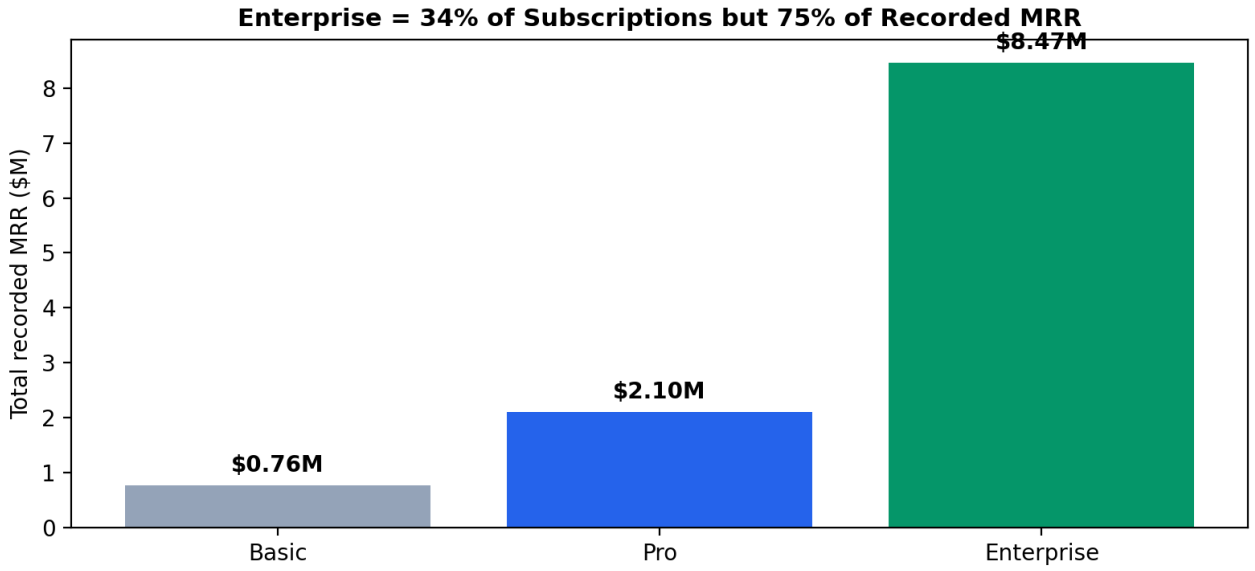


Figure 3 — Recorded MRR by plan tier (cleaned subscriptions).

Plan	Subscriptions	Recorded MRR	Avg MRR / sub	Sub churn
Basic	1,601	$760,209	$475	9.5%
Pro	1,673	$2,102,345	$1,257	9.7%
Enterprise	1,722	**$8,467,649**	$4,917	10.0%

Note on method: accounts hold multiple subscription records over time (about 10 per account), so 'recorded MRR' is the sum across subscription records, not a single month's snapshot — it is used here to compare plans, not to claim a monthly revenue figure. Upgrades appear on 10.6% of subscriptions, downgrades on 4.4% — a healthy 2.4:1 expansion ratio. Growth and churn both accelerated in 2024 as the customer base scaled (527 of 600 churn events are in 2024, alongside 4,330 new subscription records).

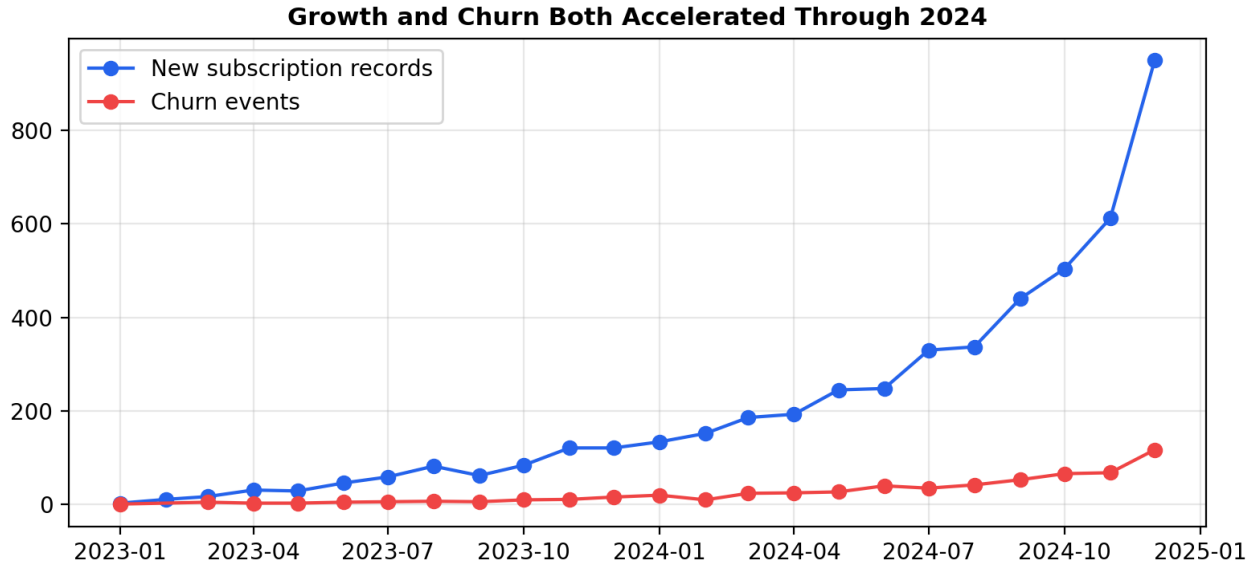


Figure 4 — Monthly new subscription records vs churn events.

3 WHY THEY LEAVE

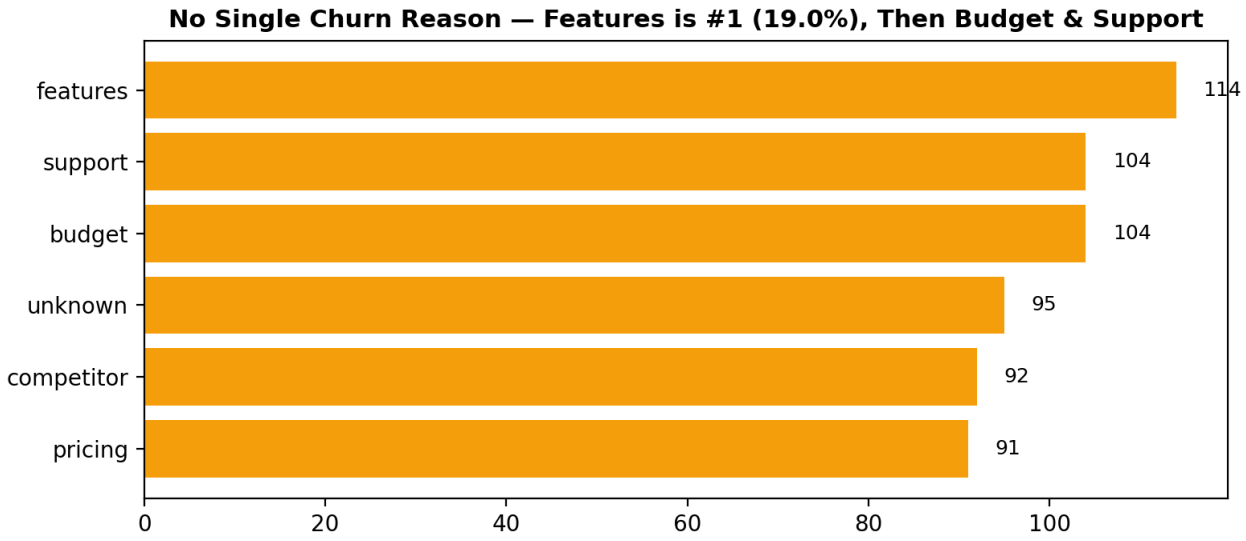


Figure 5 — Churn events by reason code (600 events).

No single reason dominates — features (19.0%), budget (17.3%) and support (17.3%) lead, with competitor and pricing at about 15% each; 15.8% are 'unknown', which itself is a data problem worth fixing with a mandatory exit question. 20.5% of churn events were preceded by an upgrade within 90 days — customers who just paid more and still left are the highest-value save targets. Total refunds are small ($8,652 across 142 events), so refunds are a symptom, not the disease.

4 PRODUCT USAGE — NO HERO FEATURE

Usage is spread almost perfectly evenly across all 40 features: the most-used feature (feature_32) accounts for only 6,686 of 250,525 total usage counts. There is no single sticky feature to build retention around — breadth, not depth, is the pattern. Beta features (10.2% of events) are **not** riskier than stable ones: 0.56 errors per event vs 0.57 for non-beta, with the same average session duration (~3,040 secs). The beta programme can be expanded without an error penalty on this evidence.

5 SUPPORT — VOLUME IS FINE, PRIORITY IS BROKEN

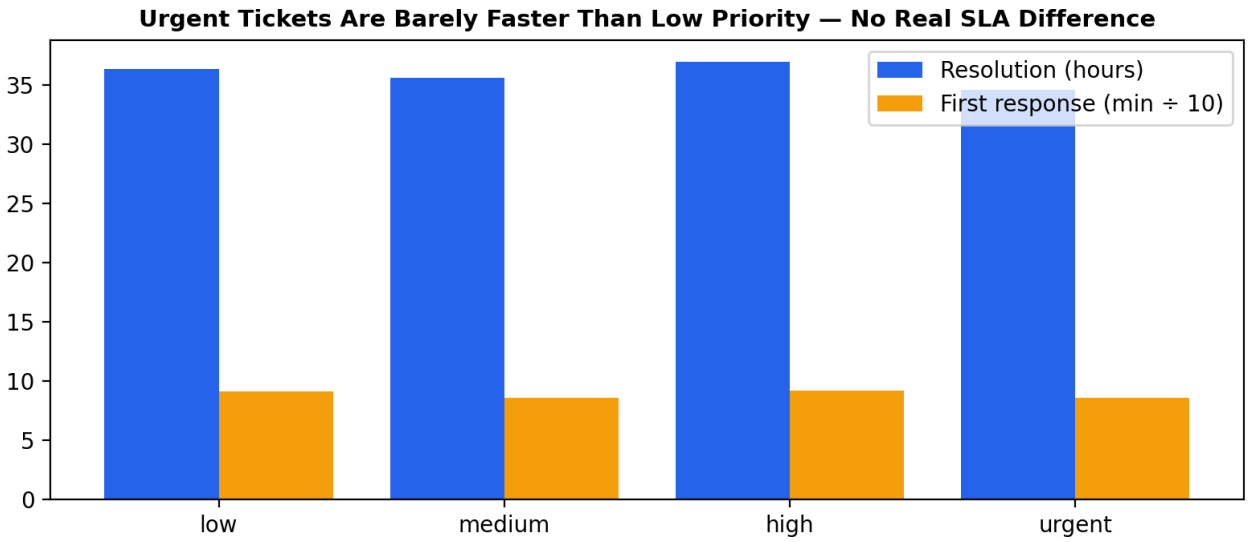


Figure 6 — Resolution time and first-response time barely change with priority.

492 of 500 accounts (98%) raised at least one ticket; average resolution is 35.9 hours and satisfaction is a healthy 3.98/5 among respondents. The failure is triage: urgent tickets resolve in 34.6 hours vs 36.4 for low priority, and first response is 86 minutes for urgent vs 91 for low. Priority labels currently change almost nothing — a real SLA risk for the Enterprise accounts that carry 75% of recorded MRR. Escalations are low (4.8%), so the queue is calm but undifferentiated.

6 COHORTS & CUSTOMER LIFETIME — REQUIRED BY THE TASK BRIEF

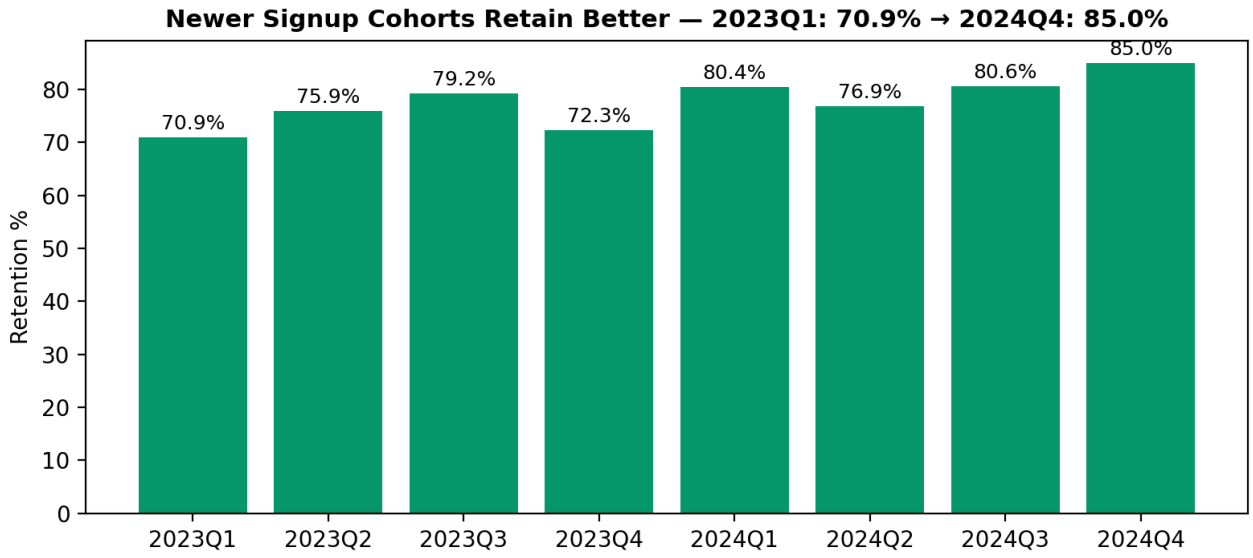


Figure 7 — Retention by signup quarter cohort. Newer cohorts retain better.

Signup cohort	Accounts	Churned	Churn	Retention
2023 Q1	55	16	29.1%	70.9%
2023 Q2	54	13	24.1%	75.9%
2023 Q3	53	11	20.8%	79.2%
2023 Q4	65	18	27.7%	72.3%
2024 Q1	56	11	19.6%	80.4%
2024 Q2	65	15	23.1%	76.9%
2024 Q3	72	14	19.4%	80.6%
2024 Q4	80	12	15.0%	**85.0%**

Cohort read: retention improves from 70.9% (2023 Q1) to 85.0% (2024 Q4). An honest caveat — newer cohorts have also had less time in which to churn, so part of the improvement is a time-window effect; the trend should be re-checked when the 2024 cohorts reach the same age as the 2023 cohorts. By plan, cohort retention is flat (Basic 78.0%, Pro 78.1%, Enterprise 77.9%) — once again, plan does not separate customers; signup timing and channel do. Monthly active subscription records grew from 3 in Jan 2023 to 4,703 in Dec 2024, so the base is expanding fast while it churns.

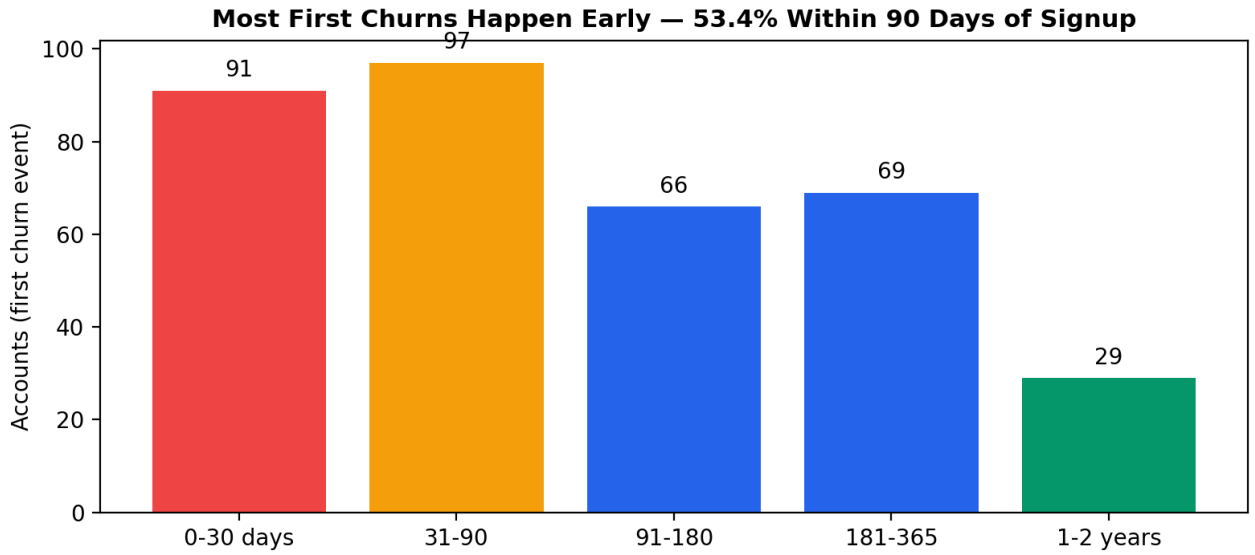


Figure 8 — Days from signup to first churn event (352 accounts). The danger zone is the first 90 days.

Customer lifetime: for accounts that ever churned, the median time to the **first** churn event is just **82 days** (mean 133 days). **53.4% of first churns happen within 90 days** of signup and 91.8% within the first year. Ended subscription records are even shorter-lived: median duration **42 days** (mean 88 days). Lifetime barely differs by plan (median 72–88 days to first churn) — customers do not leave late and loyal, they leave early and unconvinced. Retention money is therefore best spent in the first 90 days: onboarding, first-value milestones and a day-30 / day-60 check-in, not on late-stage win-back.

7 RECOMMENDATIONS

- **1. Run a DevTools retention play.** DevTools churn (31.0%) is double the best industries. Interview DevTools churners first; their top stated reasons are features and support.
- **2. Copy the partner channel, question the event channel.** Partner-acquired accounts churn at 14.6% vs 30.2% for events. Shift acquisition spend toward partners/organic and add a 90-day onboarding check-in for event sign-ups.
- **3. Protect Enterprise like it matters — because it is 75% of recorded MRR.** Named support, real urgent SLAs and quarterly reviews for Enterprise accounts; losing a handful costs more than losing dozens of Basic accounts.
- **4. Make priority mean something.** Set and enforce separate targets (e.g. urgent first response under 15 minutes, resolution under 8 hours) and report breaches weekly.
- **5. Create an upgrade-then-churn save alert.** 20.5% of churn followed a recent upgrade. Trigger a success check 30 days after every upgrade.
- **6. Fix churn measurement.** Reconcile accounts.churn_flag (110 accounts) with churn_events (352 accounts) in one definition, and make a churn reason mandatory to shrink the 15.8% 'unknown'.
- **7. Own the first 90 days.** Half of first churns happen within 90 days (median 82 days). Build a day-7 / day-30 / day-60 onboarding and check-in sequence and measure cohort retention at day 90 as the headline KPI.
- **8. Keep the duplicate checks in the pipeline.** Enforce a truly unique usage_id at generation and a uniqueness check on subscription (account, start_date, plan, MRR) so the 4 duplicate pairs found here cannot recur.

Limitations

RavenStack is a fully synthetic dataset built for learning (credit: River @ Rivalytics) — patterns are intentionally realistic but muted, which is why several classic drivers test flat. All findings describe this dataset, not a real company. Currency is USD as in the source files.

Deliverables

1) This PDF report 2) Excel dashboard: FUTURE_DS_02_RavenStack_Dashboard.xlsx 3) Source files: the 5 RavenStack CSVs + dataset README (credit to the original author preserved) 4) analysis.py to reproduce every number, including the duplicate check. Cleaning applied: 4 duplicate subscription rows removed (5,000 → 4,996); nothing else deleted.

— End of Report —

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